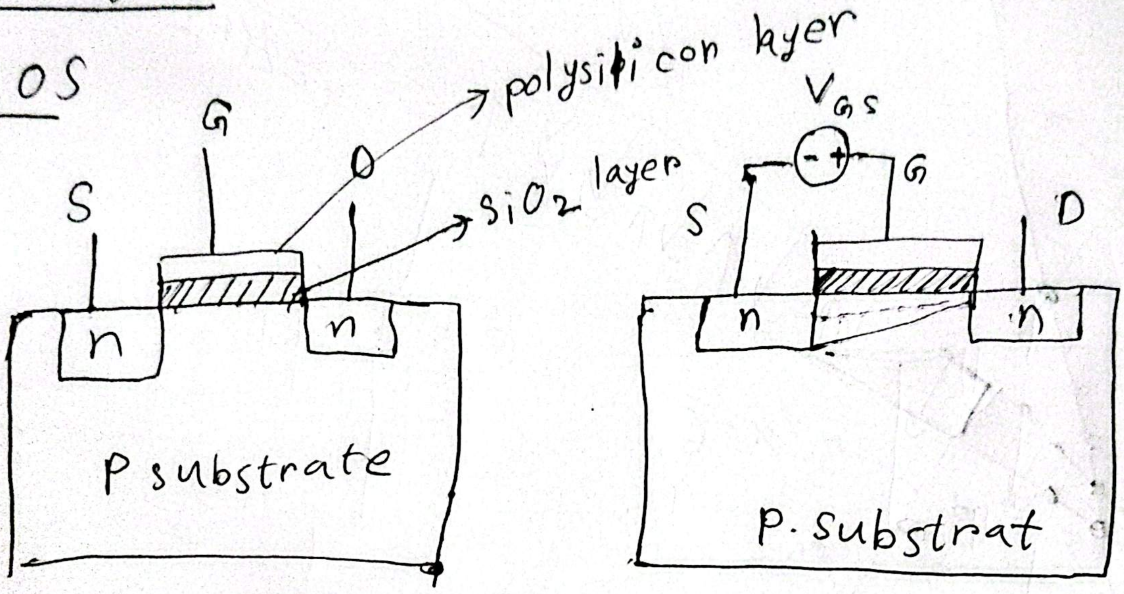
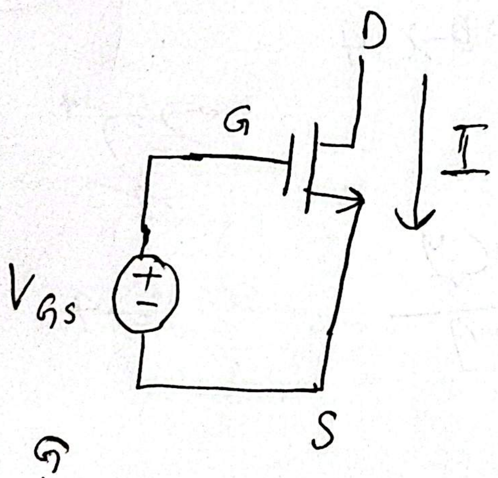
CMOS Logic

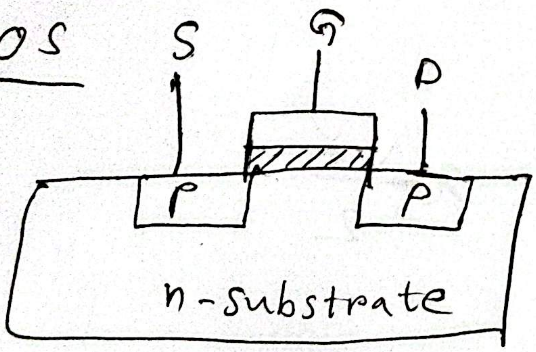
n-MOS

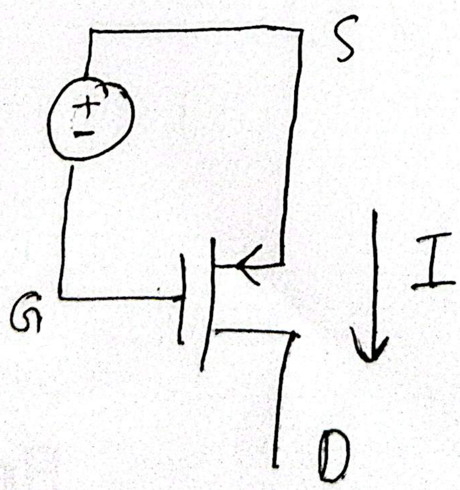




Here, $V_{GS} > V_{th}(n)$

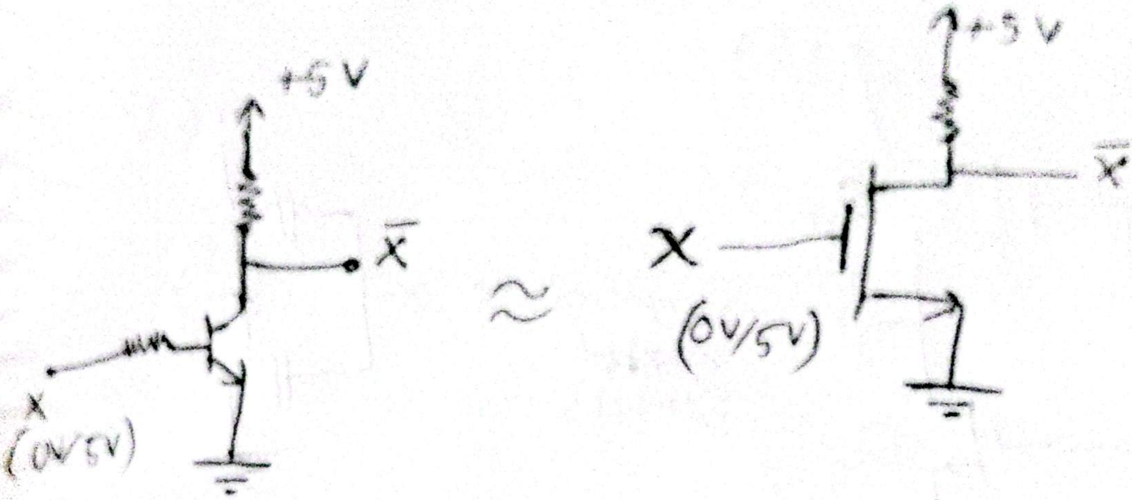
P-MOS



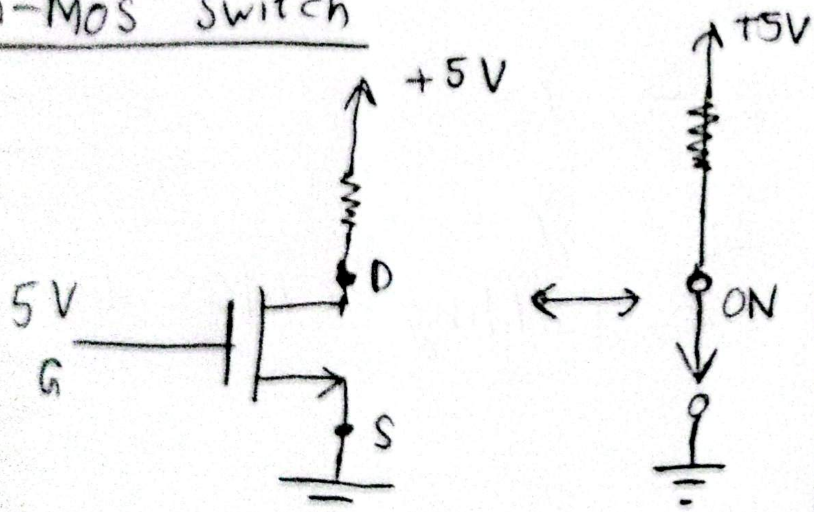


$V_{SG} > V_{th}(P)$

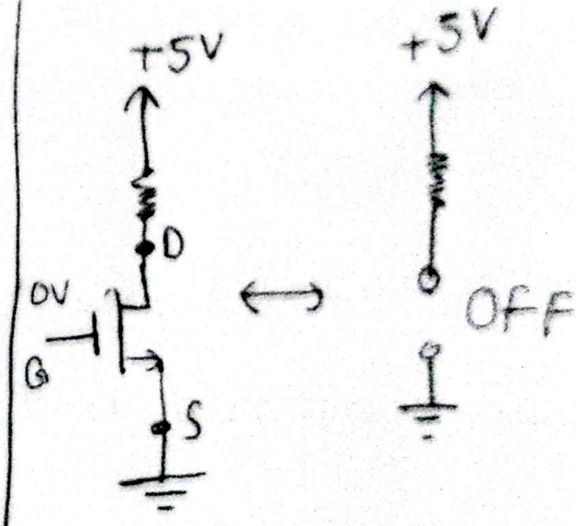
Resistor-transistor Logic (RTL) in MOSFET



n-MOS switch

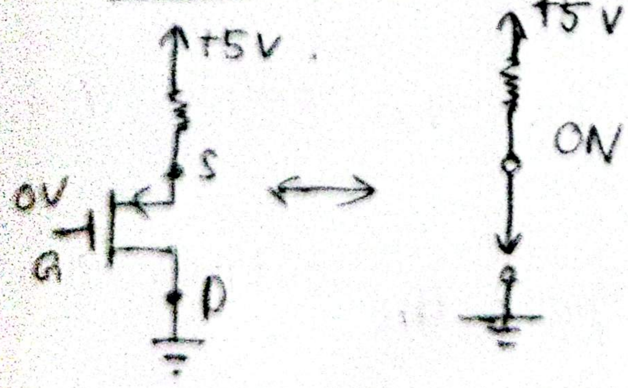


$V_{GS} = 5V - 0V = 5V$

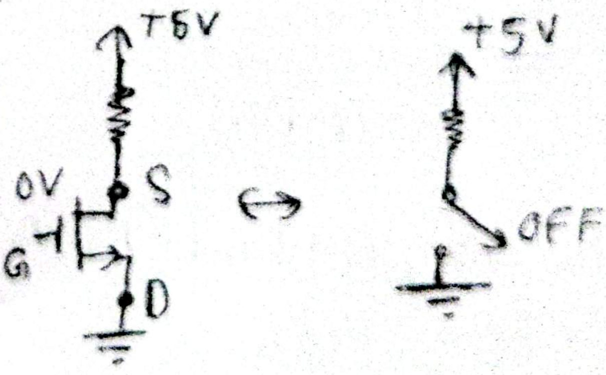


$V_{GS} = 0V - 0V = 0V$

P-MOS switch

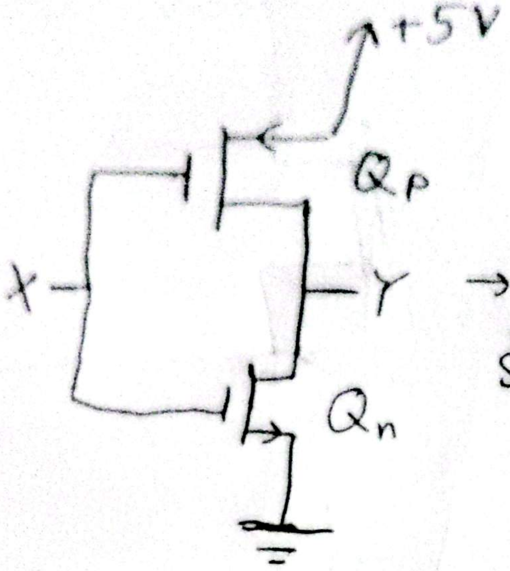


$V_{SG} = 5V - 0V = 5V$

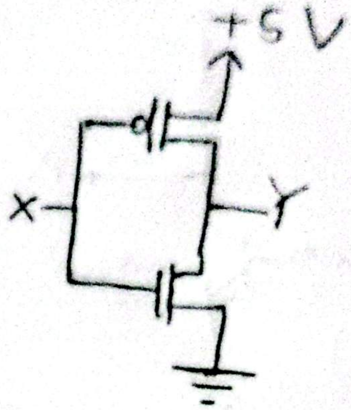


$V_{SG} = 5V - 5V = 0V$

CMOS Inverter



→ simplified → Symbols

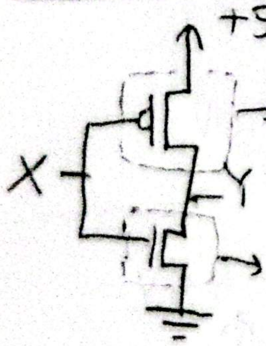


when,

$X = 0V$ $Q_p \rightarrow$ $Q_n \rightarrow$ $Y =$

When, $X = 5V$ $Q_p \rightarrow$ $Q_n \rightarrow$ $Y =$

Generalized rules for building CMOS Logic



→ Pull-up network (PUN)

→ Pulldown -network (PDN)

→ Pull-up network (PUN) pulls the output up to logic High.

→ Pull down network (PDN) pulls the output down to logic LOW.

⊛ Like CMOS Inverter, all CMOS logic can be generalized with the fact that, they consist of a PUN and a PDN.

→ PUN is built using PMOS transistors.

→ PDN is built using NMOS transistors.

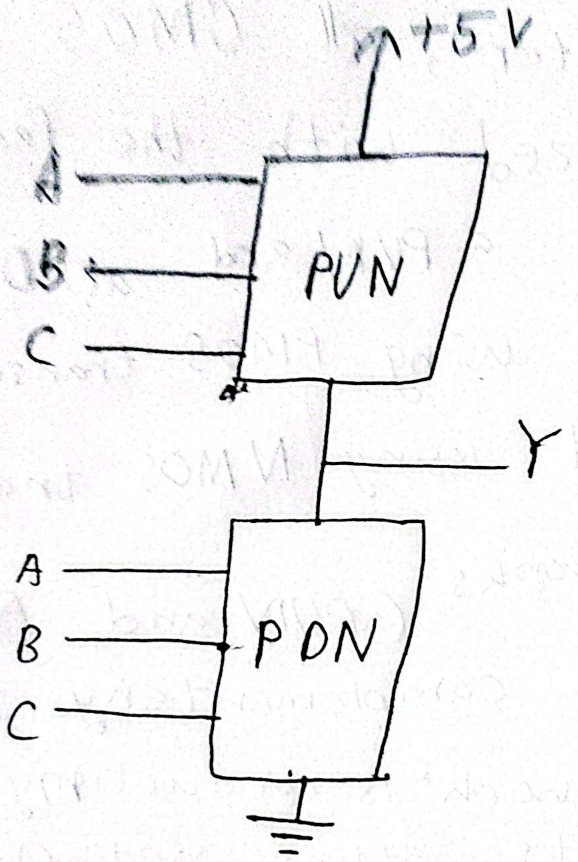
→ the two networks (PUN and PDN) functions in a complementary fashion

↳ If one network is conducting for a set of inputs, other network will be off for those inputs and vice versa.

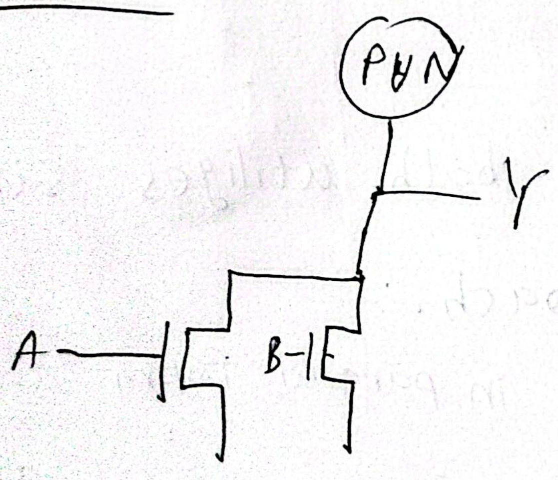
→ PDN and PUN both utiliges same construction approach.

↳ ① transistors in paralal form an OR Function.

② transistors in series form an AND function.

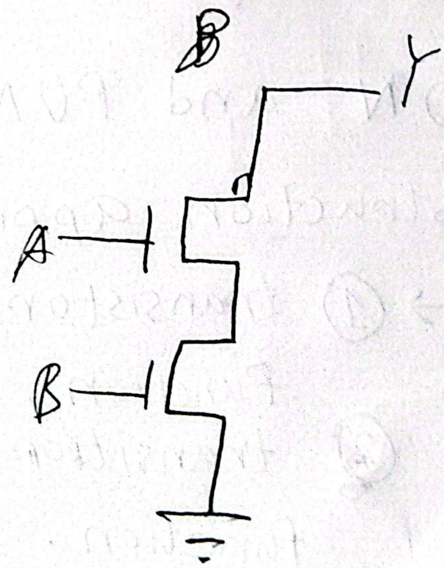


Ex 1:



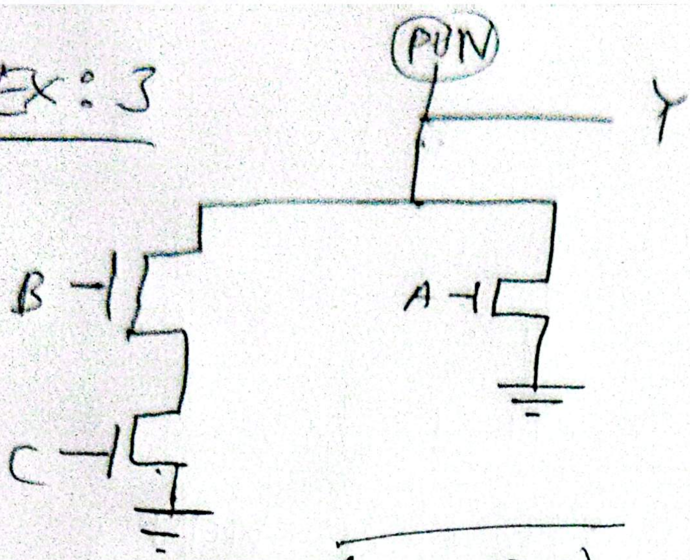
$Y = (A+B)^c$

Ex 2:



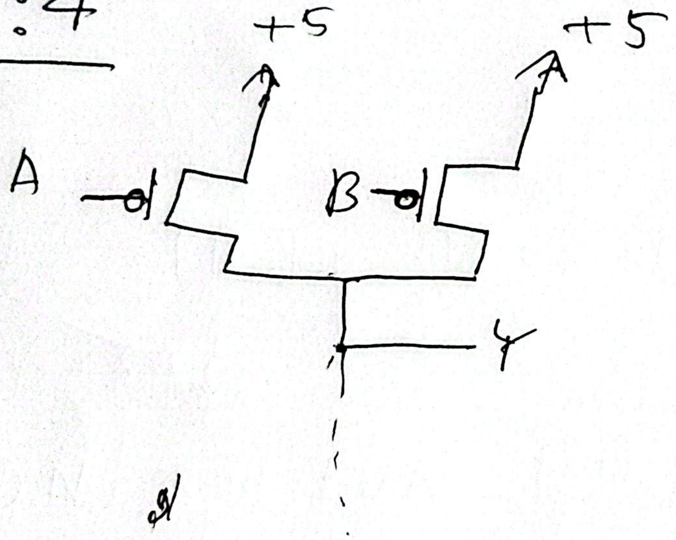
$Y = (AB)^c$

Ex: 3



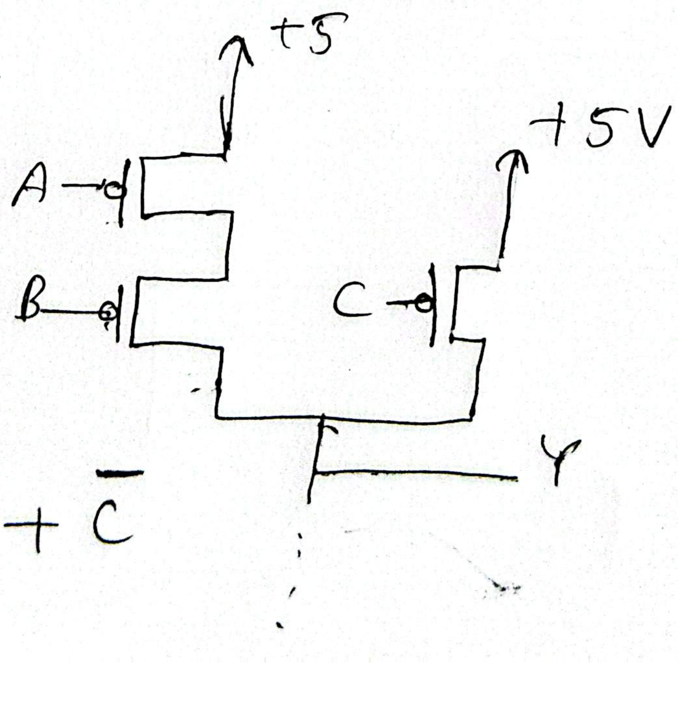
$Y = \overline{(A + BC)}$

Ex: 4



$Y = \overline{A} + \overline{B}$

Ex: 5



$Y = \overline{A} \cdot \overline{B} + \overline{C}$